

PROJECT PRESENTATION

DuoCast

Two-Person Video Calling with Local Recording & S3 Upload

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Frontend Web Development Internship · Rohit Panwar · CS-2341877

Where Real-Time Communication Meets Reliable Recording

The Project

DuoCast combines real-time peer-to-peer video communication with local browser recording and background cloud storage. The approach is designed to preserve the original capture quality even when network conditions affect the live call.



Local Recording

MediaRecorder captures each participant's own video locally.



Real-Time Calling

WebRTC provides peer-to-peer audio and video communication.



Cloud Upload

Recording chunks are uploaded to AWS S3 in the background.

Project at a Glance

Core technologies and project focus

1

Position

DuoCast

Two-person video calling
application

2

Location

Web Browser

Client-side real-time
communication

3

Duration

Internship

Frontend Web Development
project

4

Resources

Next.js + React

Modern frontend
application stack

How DuoCast Works

From joining a room to storing the final recording



The Challenge We Address

Why local recording matters for real-time video

1

Network Quality

Poor bandwidth or packet loss can cause choppy or degraded live video.

2

Stream Recording

Recording the remote WebRTC stream can preserve freezes, lag and resolution drops.

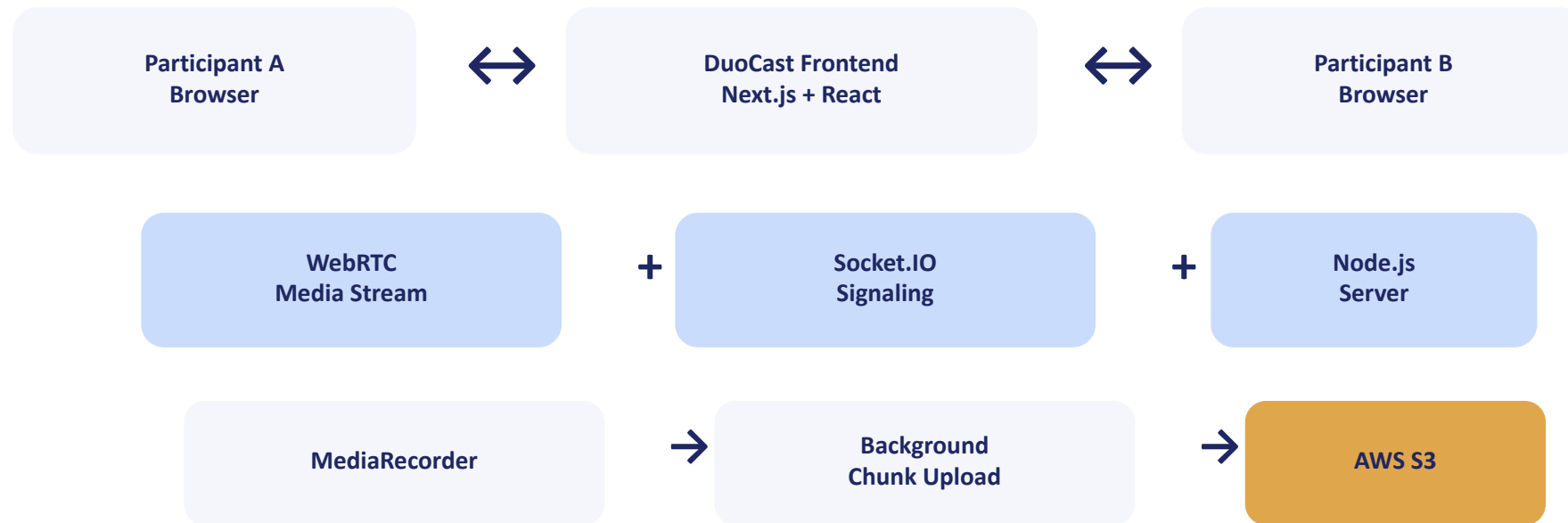
3

Quality Loss

A weak connection on one side can negatively affect the resulting recording.

System Architecture

Client-side architecture with WebRTC signaling, local recording and cloud storage.



Optional / experimental: FFmpeg-based side-by-side merge of participant recordings.

Development Journey

From interface design to recording and cloud storage

1

Interface & Call Setup

Built the web interface and established two-person video communication using WebRTC.

2

Recording & Storage

Implemented local browser recording and background chunk uploads to AWS S3.

3

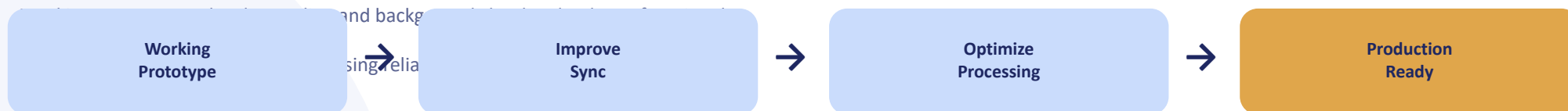
Synchronization & Processing

Added signaling controls for synchronized recording and a workflow for combining recordings.

What Comes Next

Congratulations on your selection. Should you have any questions before or during the program, feel free to reach out at your earliest convenience.

Current State



Thank You

Thank you for your time and consideration.

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Questions?